EXERCISE
(OBJECTIVE TYPE QUESTIONS)

Directions: *Mark (▯) against the correct answer in each of the following:*

1. What is the place value of 5 in 3254710? **(CLAT, 2010)**
(*a*) 5 (*b*) 10000
(*c*) 50000 (*d*) 54710

2. The face value of 8 in the number 458926 is **(R.R.B., 2006)**
(*a*) 8 (*b*) 1000
(*c*) 8000 (*d*) 8926

3. The sum of the place values of 3 in the number 503535 is **(M.B.A., 2005)**
(*a*) 6 (*b*) 60
(*c*) 3030 (*d*) 3300

4. The difference between the place values of 7 and 3 in the number 527435 is
(*a*) 4 (*b*) 5
(*c*) 45 (*d*) 6970

5. The difference between the local value and the face value of 7 in the numeral 32675149 is
(*a*) 5149 (*b*) 64851
(*c*) 69993 (*d*) 75142
(*e*) None of these

6. The sum of the greatest and smallest number of five digits is **(M.C.A., 2005)**
(*a*) 11,110 (*b*) 10,999
(*c*) 109,999 (*d*) 111,110

7. If the largest three-digit number is subtracted from the smallest five-digit number, then the remainder is
(*a*) 1 (*b*) 9000
(*c*) 9001 (*d*) 90001

8. The smallest number of 5 digits beginning with 3 and ending with 5 will be **(R.R.B., 2006)**
(*a*) 31005 (*b*) 30015
(*c*) 30005 (*d*) 30025

9. What is the minimum number of four digits formed by using the digits 2, 4, 0, 7? **(P.C.S., 2007)**
(*a*) 2047 (*b*) 2247
(*c*) 2407 (*d*) 2470

10. All natural numbers and 0 are called the ..... numbers. **(R.R.B., 2006)**
(*a*) rational (*b*) integer
(*c*) whole (*d*) prime

11. Consider the following statements about natural numbers:
(1) There exists a smallest natural number.
(2) There exists a largest natural number.
(3) Between two natural numbers, there is always a natural number.
Which of the above statements is/are correct?
(*a*) None (*b*) Only 1
(*c*) 1 and 2 (*d*) 2 and 3

12. Every rational number is also
(*a*) an integer (*b*) a real number
(*c*) a natural number (*d*) a whole number

13. The number π is **(R.R.B., 2005)**
(*a*) a fraction (*b*) a recurring decimal
(*c*) a rational number (*d*) an irrational number

14. $\sqrt{2}$ is a/an
(*a*) rational number (*b*) natural number
(*c*) irrational number (*d*) integer

15. The number $\sqrt{3}$ is
(*a*) a finite decimal
(*b*) an infinite recurring decimal
(*c*) equal to 1.732
(*d*) an infinite non-recurring decimal

16. There are just two ways in which 5 may be expressed as the sum of two different positive (non-zero) integers, namely 5 = 4 + 1 = 3 + 2. In how many ways, 9 can be expressed as the sum of two different positive (non-zero) integers?
(*a*) 3 (*b*) 4
(*c*) 5 (*d*) 6

17. P and Q are two positive integers such that $PQ = 64$. Which of the following cannot be the value of $P + Q$?
(*a*) 16 (*b*) 20
(*c*) 35 (*d*) 65

18. If $x + y + z = 9$ and both y and z are positive integers greater than zero, then the maximum value x can take is **(Campus Recruitment, 2006)**
(*a*) 3 (*b*) 7
(*c*) 8 (*d*) Data insufficient

19. What is the sum of the squares of the digits from 1 to 9?
(*a*) 105 (*b*) 260
(*c*) 285 (*d*) 385

20. If n is an integer between 20 and 80, then any of the following could be $n + 7$ except
(*a*) 47 (*b*) 58
(*c*) 84 (*d*) 88

21. Which one of the following is the correct sequence in respect of the Roman numerals: C, D, L and M? **(Civil Services, 2008)**
(*a*) $C > D > L > M$ (*b*) $M > L > D > C$
(*c*) $M > D > C > L$ (*d*) $L > C > D > M$

22. If the numbers from 1 to 24, which are divisible by 2 are arranged in descending order, which number will be at the 8th place from the bottom? **(CLAT, 2010)**

(*a*) 10 (*b*) 12
(*c*) 16 (*d*) 18

23. 2 – 2 + 2 – 2 + ... 101 terms =? **(P.C.S., 2008)**
(*a*) – 2 (*b*) 0
(*c*) 2 (*d*) None of these

24. 98th term of the infinite series 1, 2, 3, 4, 1, 2, 3, 4, 1, 2, ... is **(M.C.A., 2005)**
(*a*) 1 (*b*) 2
(*c*) 3 (*d*) 4

25. If x, y, z be the digits of a number beginning from the left, the number is
(*a*) xyz (*b*) $x + 10y + 100z$
(*c*) $10x + y + 100z$ (*d*) $100x + 10y + z$

26. If x, y, z and w be the digits of a number beginning from the left, the number is
(*a*) $xyzw$
(*b*) $wzyx$
(*c*) $x + 10y + 100z + 1000w$
(*d*) $10^3x + 10^2y + 10z + w$

27. If n and p are both odd numbers, which of the following is an even number?
(*a*) $n + p$ (*b*) $n + p + 1$
(*c*) $np + 2$ (*d*) np

28. For the integer n, if n^3 is odd, then which of the following statements are true?
I. n is odd. II. n^2 is odd.
III. n^2 is even.
(*a*) I only (*b*) II only
(*c*) I and II only (*d*) I and III only

29. If $(n - 1)$ is an odd number, what are the two other odd numbers nearest to it?
(*a*) $n, n - 1$ (*b*) $n, n - 2$
(*c*) $n - 3, n + 1$ (*d*) $n - 3, n + 5$

30. Which of the following is always odd?
(*a*) Sum of two odd numbers
(*b*) Difference of two odd numbers
(*c*) Product of two odd numbers
(*d*) None of these

31. If x is an odd integer, then which of the following is true?
(*a*) $5x - 2$ is even (*b*) $5x^2 + 2$ is odd
(*c*) $5x^2 + 3$ is odd (*d*) None of these

32. If a and b are two numbers such that $ab = 0$, then **(R.R.B., 2006)**
(*a*) $a = 0$ and $b = 0$ (*b*) $a = 0$ or $b = 0$ or both
(*c*) $a = 0$ and $b \neq 0$ (*d*) $b = 0$ and $a \neq 0$

33. If A, B, C, D are numbers in increasing order and D, B, E are numbers in decreasing order, then which one of the following sequences need neither be in a decreasing nor in an increasing order?
(*a*) E, C, D (*b*) E, B, C
(*c*) D, B, A (*d*) A, E, C

34. If m, n, o, p and q are integers, then $m(n + o)(p - q)$ must be even when which of the following is even?
(*a*) m (*b*) p
(*c*) $m + n$ (*d*) $n + p$

35. If n is a negative number, then which of the following is the least?
(*a*) 0 (*b*) $-n$
(*c*) $2n$ (*d*) n^2

36. If $x - y = 8$, then which of the following must be true?
I. Both x and y are positive.
II. If x is positive, y must be positive.
III. If x is negative, y must be negative.
(*a*) I only (*b*) II only
(*c*) I and II (*d*) III only

37. If x and y are negative, then which of the following statements is/are always true?
I. $x + y$ is positive.
II. xy is positive.
III. $x - y$ is positive.
(*a*) I only (*b*) II only
(*c*) III only (*d*) I and III only

38. If $n = 1 + x$, where x is the product of four consecutive positive integers, then which of the following is/are true?
I. n is odd. II. n is prime.
III. n is a perfect square.
(*a*) I only (*b*) I and II only
(*c*) I and III only (*d*) None of these

39. If $x = \frac{2}{5}y + 3$, how does y change when x increases from 1 to 2?
(*a*) y increases from -5 to $-\frac{5}{2}$
(*b*) y increases from $\frac{2}{5}$ to 5
(*c*) y increases from $\frac{5}{2}$ to 5
(*d*) y decreases from -5 to $-\frac{5}{2}$

40. If x is a rational number and y is an irrational number, then
(*a*) both $x + y$ and xy are necessarily rational
(*b*) both $x + y$ and xy are necessarily irrational
(*c*) xy is necessarily irrational, but $x + y$ can be either rational or irrational
(*d*) $x + y$ is necessarily irrational, but xy can be either rational or irrational

41. The difference between the square of any two consecutive integers is equal to
(*a*) sum of two numbers
(*b*) difference of two numbers
(*c*) an even number
(*d*) product of two numbers

42. Between two distinct rational numbers a and b, there exists another rational number which is **(P.C.S., 2006)**

(*a*) $\frac{a}{2}$ (*b*) $\frac{b}{2}$

(*c*) $\frac{ab}{2}$ (*d*) $\frac{a+b}{2}$

43. If $B > A$, then which expression will have the highest value (given that A and B are positive integers)? **(Campus Recruitment, 2007)**

(*a*) $A - B$ (*b*) AB

(*c*) $A + B$ (*d*) Can't say

44. If $0 < x < 1$, which of the following is greatest? **(Campus Recruitment, 2007)**

(*a*) x (*b*) x^2

(*c*) $\frac{1}{x}$ (*d*) $\frac{1}{x^2}$

45. If p is a positive fraction less than 1, then

(*a*) $\frac{1}{p}$ is less than 1 (*b*) $\frac{1}{p}$ is a positive integer

(*c*) p^2 is less than p

(*d*) $\frac{2}{p} - p$ is a positive number

46. If x is a real number, then $x^2 + x + 1$ is

(*a*) less than $\frac{3}{4}$

(*b*) zero for at least one value of x

(*c*) always negative

(*d*) greater than or equal to $\frac{3}{4}$

47. Let n be a natural number such that $\frac{1}{2}+\frac{1}{3}+\frac{1}{7}+\frac{1}{n}$ is also a natural number. Which of the following statements is not true? **(A.A.O. Exam, 2009)**

(*a*) 2 divides n (*b*) 3 divides n

(*c*) 7 divides n (*d*) $n > 84$

48. If n is an integer, how many values of n will give an integral value of $\left(\frac{16n^2+7n+6}{n}\right)$?

(*a*) 2 (*b*) 3

(*c*) 4 (*d*) None of these

49. If $p > q$ and $r < 0$, then which is true?

(*a*) $pr < qr$ (*b*) $p - r < q - r$

(*c*) $p + r < q + r$ (*d*) None of these

50. If $X < Z$ and $X < Y$, which of the following is necessarily true?

I. $Y < Z$ II. $X^2 < YZ$

III. $ZX < Y + Z$

(*a*) Only I (*b*) Only II

(*c*) Only III (*d*) None of these

51. In the relation $x > y + z$, $x + y > p$ and $z < p$, which of the following is necessarily true? **(Campus Recruitment, 2008)**

(*a*) $y > p$ (*b*) $x + y > z$

(*c*) $y + p > x$ (*d*) Insufficient data

52. If a and b are positive integers and $\frac{(a-b)}{3.5} = \frac{4}{7}$, then **(Campus Recruitment, 2010)**

(*a*) $b > a$ (*b*) $b < a$

(*c*) $b = a$ (*d*) $b \geq a$

53. If $13 = \frac{13\,w}{(1-w)}$, then $(2w)^2 = ?$ **(Campus Recruitment, 2009)**

(*a*) $\frac{1}{4}$ (*b*) $\frac{1}{2}$

(*c*) 1 (*d*) 2

Directions (Questions 54–57): *For a 5–digit number, without repetition of digits, the following information is available.* **(B.B.A., 2006)**

(*i*) The first digit is more than 5 times the last digit.

(*ii*) The two-digit number formed by the last two digits is the product of two prime numbers.

(*iii*) The first three digits are all odd.

(*iv*) The number does not contain the digits 3 or 0 and the first digit is also the largest.

54. The second digit of the number is

(*a*) 5 (*b*) 7

(*c*) 9

(*d*) Cannot be determined

55. The last digit of the number is

(*a*) 0 (*b*) 1

(*c*) 2 (*d*) 3

56. The largest digit in the number is

(*a*) 5 (*b*) 7

(*c*) 8 (*d*) 9

57. Which of the following is a factor of the given number?

(*a*) 2 (*b*) 3

(*c*) 4 (*d*) 9

58. The least prime number is

(*a*) 0 (*b*) 1

(*c*) 2 (*d*) 3

59. Consider the following statements:

1. If x and y are composite numbers, then $x + y$ is always composite.
2. There does not exist a natural number which is neither prime nor composite.

Which of the above statements is/are correct?

(*a*) 1 only (*b*) 2 only

(*c*) Both 1 and 2 (*d*) Neither 1 nor 2

60. The number of prime numbers between 0 and 50 is

(*a*) 14 (*b*) 15

(*c*) 16 (*d*) 17

61. The prime numbers dividing 143 and leaving a remainder of 3 in each case are
(*a*) 2 and 11 (*b*) 11 and 13
(*c*) 3 and 7 (*d*) 5 and 7

62. The sum of the first four primes is
(*a*) 10 (*b*) 11
(*c*) 16 (*d*) 17

63. The sum of all the prime numbers from 1 to 20 is
(*a*) 75 (*b*) 76
(*c*) 77 (*d*) 78

64. A prime number N, in the range 10 to 50, remains unchanged when its digits are reversed. The square of such a number is
(*a*) 121 (*b*) 484
(*c*) 1089 (*d*) 1936

65. The remainder obtained when any prime number greater than 6 is divided by 6 must be **(Campus Recruitment, 2007)**
(*a*) either 1 or 2 (*b*) either 1 or 3
(*c*) either 1 or 5 (*d*) either 3 or 5

66. Which of the following is not a prime number? **(CLAT, 2010)**
(*a*) 21 (*b*) 23
(*c*) 29 (*d*) 43

67. Which of the following is a prime number? **(CLAT, 2010)**
(*a*) 19 (*b*) 20
(*c*) 21 (*d*) 22

68. Which of the following is a prime number? **(Campus Recruitment, 2008)**
(*a*) 115 (*b*) 119
(*c*) 127 (*d*) None of these

69. Which of the following is a prime number? **(R.R.B., 2006)**
(*a*) 143 (*b*) 289
(*c*) 117 (*d*) 359

70. The smallest value of natural number n, for which $2n + 1$ is not a prime number, is
(*a*) 3 (*b*) 4
(*c*) 5 (*d*) None of these

71. The smallest three-digit prime number is
(*a*) 101 (*b*) 103
(*c*) 107 (*d*) None of these

72. How many of the integers between 110 and 120 are prime numbers? **(M.B.A., 2006)**
(*a*) 0 (*b*) 1
(*c*) 2 (*d*) 3
(*e*) 4

73. Four prime numbers are arranged in ascending order. The product of first three is 385 and that of last three is 1001. The largest prime number is **(R.R.B., 2006)**
(*a*) 9 (*b*) 11
(*c*) 13 (*d*) 17

74. The sum of three prime numbers is 100. If one of them exceeds another by 36, then one of the numbers is
(*a*) 7 (*b*) 29
(*c*) 41 (*d*) 67

75. Which one of the following is a prime number?
(*a*) 161 (*b*) 221
(*c*) 373 (*d*) 437

76. The smallest prime number, that is the fifth term of an increasing arithmetic sequence in which all the four preceding terms are also prime, is
(*a*) 17 (*b*) 29
(*c*) 37 (*d*) 53

77. The number of prime numbers between 301 and 320 are
(*a*) 3 (*b*) 4
(*c*) 5 (*d*) 6

78. Consider the following statements:
1. If $p > 2$ is a prime, then it can be written as $4n + 1$ or $4n + 3$ for a suitable natural number n.
2. If $p > 2$ is a prime, then $(p - 1)(p + 1)$ is always divisible by 4.

Of these statements,
(*a*) (1) is true but (2) is false
(*b*) (1) is false but (2) is true
(*c*) (1) and (2) are false
(*d*) (1) and (2) are true

79. What is the first value of n for which $n^2 + n + 41$ is not a prime?
(*a*) 1 (*b*) 10
(*c*) 20 (*d*) 40

80. Let $X_k = (p_1 p_2 \ldots\ldots p_k) + 1$, where $p_1, p_2, \ldots\ldots\ldots, p_k$ are the first k primes.
Consider the following:
1. X_k is a prime number.
2. X_k is a composite number.
3. $X_k + 1$ is always an even number.

Which of the above is/are correct?
(*a*) 1 only (*b*) 2 only
(*c*) 3 only (*d*) 1 and 3

81. $6 \times 3 (3 - 1)$ is equal to **(CLAT, 2010)**
(*a*) 19 (*b*) 20
(*c*) 36 (*d*) 53

82. $1234 + 2345 - 3456 + 4567 = ?$ **(Bank Recruitment, 2010)**
(*a*) 4590 (*b*) 4670
(*c*) 4680 (*d*) 4690
(*e*) None of these

83. $5566 - 7788 + 9988 = ? + 4444$ **(Bank Recruitment, 2010)**
(*a*) 3223 (*b*) 3232
(*c*) 3322 (*d*) 3333
(*e*) None of these

84. 38649 – 1624 – 4483 =? **(Bank Recruitment, 2009)**
(*a*) 32425 (*b*) 32452
(*c*) 34522 (*d*) 35422
(*e*) None of these

85. 884697 – 773697 – 102479 =? **(Bank Recruitment, 2009)**
(*a*) 8251 (*b*) 8512
(*c*) 8521 (*d*) 8531
(*e*) None of these

86. 10531 + 4813 – 728 =? × 87 **(Bank Recruitment, 2008)**
(*a*) 168 (*b*) 172
(*c*) 186 (*d*) 212
(*e*) None of these

87. What is 394 times 113?
(*a*) 44402 (*b*) 44522
(*c*) 44632 (*d*) 44802
(*e*) None of these

88. 1260 ÷ 14 ÷ 9 =? **(Bank P.O., 2009)**
(*a*) 9 (*b*) 10
(*c*) 81 (*d*) 810
(*e*) None of these

89. 136 × 12 × 8 =? **(Bank P.O., 2009)**
(*a*) 12066 (*b*) 13046
(*c*) 13064 (*d*) 13066
(*e*) None of these

90. 8888 + 848 + 88 –? = 7337 + 737 **(Bank P.O., 2009)**
(*a*) 1450 (*b*) 1550
(*c*) 1650 (*d*) 1750
(*e*) None of these

91. 414 ×? × 7 = 127512 **(Bank P.O., 2009)**
(*a*) 36 (*b*) 40
(*c*) 44 (*d*) 48
(*e*) None of these

92. Product of 82540027 and 43253 is
(*a*) 3570103787831 (*b*) 3570103787832
(*c*) 3570103787833 (*d*) 3570103787834

93. (46351 – 36418 – 4505) ÷? = 1357 **(Bank P.O., 2009)**
(*a*) 2 (*b*) 3
(*c*) 4 (*d*) 6
(*e*) None of these

94. 6 × 66 × 666 =? **(Bank Recruitment, 2007)**
(*a*) 263376 (*b*) 263763
(*c*) 263736 (*d*) 267336
(*e*) None of these

95. If you subtract – 1 from + 1, what will be the result? **(R.R.B., 2006)**
(*a*) – 2 (*b*) 0
(*c*) 1 (*d*) 2

96. 8 + 88 + 888 + 8888 + 88888 + 888888 =?
(*a*) 897648 (*b*) 896748
(*c*) 986748 (*d*) 987648
(*e*) None of these

97. From the sum of 17 and – 12, subtract 48. **(E.S.I.C., 2006)**
(*a*) – 43 (*b*) – 48
(*c*) – 17 (*d*) – 20

98. 60840 ÷ 234 =?
(*a*) 225 (*b*) 255
(*c*) 260 (*d*) 310
(*e*) None of these

99. 3578 + 5729 –? × 581 = 5821
(*a*) 3 (*b*) 4
(*c*) 6 (*d*) None of these

100. – 95 ÷ 19 =?
(*a*) – 5 (*b*) – 4
(*c*) 0 (*d*) 5

101. 12345679 × 72 is equal to
(*a*) 88888888 (*b*) 888888888
(*c*) 898989898 (*d*) 999999998

102. 8899 – 6644 – 3322 =? – 1122
(*a*) 55 (*b*) 65
(*c*) 75 (*d*) 85
(*e*) None of these

103. 74844 ÷? = 54 × 63 **(Bank P.O., 2009)**
(*a*) 22 (*b*) 34
(*c*) 42 (*d*) 54
(*e*) None of these

104. 1256 × 3892 =?
(*a*) 4883852 (*b*) 4888532
(*c*) 4888352 (*d*) 4883582
(*e*) None of these

105. What is 786 times 964? **(Bank P.O., 2008)**
(*a*) 757704 (*b*) 754164
(*c*) 759276 (*d*) 749844
(*e*) None of these

106. What is 348 times 265? **(S.B.I.P.O., 2008)**
(*a*) 88740 (*b*) 89750
(*c*) 92220 (*d*) 95700
(*e*) None of these

107. (71 × 29 + 27 × 15 + 8 × 4) equals **(S.S.C., 2007)**
(*a*) 2496 (*b*) 3450
(*c*) 3458 (*d*) None of these

108. ? × (|*a*| × |*b*|) = – *ab*
(*a*) 0 (*b*) –1
(*c*) 1 (*d*) None of these

109. $(46)^2 - (?)^2 = 4398 - 3066$
(*a*) 16 (*b*) 28
(*c*) 36 (*d*) 42
(*e*) None of these

110. (800 ÷ 64) × (1296 ÷ 36) =?
(*a*) 420 (*b*) 460
(*c*) 500 (*d*) 540
(*e*) None of these

111. 5358 × 51 =?
(*a*) 273258 (*b*) 273268
(*c*) 273348 (*d*) 273358

112. 587 × 999 =?
(*a*) 586413 (*b*) 587523
(*c*) 614823 (*d*) 615173

113. 3897 × 999 =?
(*a*) 3883203 (*b*) 3893103
(*c*) 3639403 (*d*) 3791203
(*e*) None of these

114. 72519 × 9999 =?
(*a*) 725117481 (*b*) 674217481
(*c*) 685126481 (*d*) 696217481
(*e*) None of these

115. 2056 × 987 =?
(*a*) 1936372 (*b*) 2029272
(*c*) 1896172 (*d*) 1923472
(*e*) None of these

116. 1904 × 1904 =?
(*a*) 3654316 (*b*) 3632646
(*c*) 3625216 (*d*) 3623436
(*e*) None of these

117. 1397 × 1397 =?
(*a*) 1951609 (*b*) 1981709
(*c*) 18362619 (*d*) 2031719
(*e*) None of these

118. 107 × 107 + 93 × 93 =?
(*a*) 19578 (*b*) 19418
(*c*) 20098 (*d*) 21908
(*e*) None of these

119. 217 × 217 + 183 × 183 =? **(R.R.B., 2007)**
(*a*) 79698 (*b*) 80578
(*c*) 80698 (*d*) 81268
(*e*) None of these

120. 106 × 106 − 94 × 94 =?
(*a*) 2400 (*b*) 2000
(*c*) 1904 (*d*) 1906
(*e*) None of these

121. 8796 × 223 + 8796 × 77 =?
(*a*) 2736900 (*b*) 2738800
(*c*) 2658560 (*d*) 2716740
(*e*) None of these

122. 287 × 287 + 269 × 269 − 2 × 287 × 269 =?
(*a*) 534 (*b*) 446
(*c*) 354 (*d*) 324
(*e*) None of these

123. $\{(476 + 424)^2 - 4 \times 476 \times 424\}$ =?
(*a*) 2906 (*b*) 3116
(*c*) 2704 (*d*) 2904
(*e*) None of these

124. The value of 112×5^4 is
(*a*) 6700 (*b*) 70000
(*c*) 76500 (*d*) 77200

125. Multiply 5746320819 by 125.
(*a*) 718,290,102,375 (*b*) 728,490,301,375
(*c*) 748,290,103,375 (*d*) 798,290,102,975

126. 935421 × 625 =?
(*a*) 575648125 (*b*) 584638125
(*c*) 584649125 (*d*) 585628125

127. $(999)^2 - (998)^2$ =? **(R.R.B., 2008)**
(*a*) 1992 (*b*) 1995
(*c*) 1997 (*d*) 1998

128. $(80)^2 - (65)^2 + 81$ =?
(*a*) 306 (*b*) 2094
(*c*) 2175 (*d*) 2256
(*e*) None of these

129. $(24 + 25 + 26)^2 - (10 + 20 + 25)^2$ =?
(*a*) 352 (*b*) 400
(*c*) 752 (*d*) 2600
(*e*) None of these

130. $(65)^2 - (55)^2$ =?
(*a*) 10 (*b*) 100
(*c*) 120 (*d*) 1200

131. If a and b be positive integers such that $a^2 - b^2 = 19$, then the value of a is **(S.S.C., 2010)**
(*a*) 9 (*b*) 10
(*c*) 19 (*d*) 20

132. If a and b are positive integers, $a > b$ and $(a + b)^2 - (a - b)^2 > 29$, then the smallest value of a is
(*a*) 3 (*b*) 4
(*c*) 6 (*d*) 7

133. 397 × 397 + 104 × 104 + 2 × 397 × 104 =?
(*a*) 250001 (*b*) 251001
(*c*) 260101 (*d*) 261001

134. If $(64)^2 - (36)^2 = 20 \times x$, then x =?
(*a*) 70 (*b*) 120
(*c*) 180 (*d*) 140
(*e*) None of these

135. $\dfrac{(489 + 375)^2 - (489 - 375)^2}{(489 \times 375)} = ?$
(*a*) 144 (*b*) 864
(*c*) 2 (*d*) 4
(*e*) None of these

136. $\dfrac{(963 + 476)^2 + (963 - 476)^2}{(963 \times 963 + 476 \times 476)} = ?$
(*a*) 2 (*b*) 4

(*c*) 497 (*d*) 1449
(*e*) None of these

137. $\dfrac{768\times768\times768+232\times232\times232}{786\times768-768\times232+232\times232}=?$
(*a*) 1000 (*b*) 536
(*c*) 500 (*d*) 268
(*e*) None of these

138. $\dfrac{854\times854\times854-276\times276\times276}{854\times854+854\times276+276\times276}=?$
(*a*) 1130 (*b*) 578
(*c*) 565 (*d*) 1156
(*e*) None of these

139. $\dfrac{753\times753+247\times247-753\times247}{753\times753\times753+247\times247\times247}=?$
(*a*) $\dfrac{1}{1000}$ (*b*) $\dfrac{1}{506}$
(*c*) $\dfrac{253}{500}$ (*d*) None of these

140. $\dfrac{256\times256-144\times144}{112}$ is equal to **(S.S.C., 2010)**
(*a*) 420 (*b*) 400
(*c*) 360 (*d*) 320

141. If $a = 11$ and $b = 9$, then the value of $\left(\dfrac{a^2+b^2+ab}{a^3-b^3}\right)$ is **(S.S.C., 2010)**
(*a*) $\dfrac{1}{2}$ (*b*) $\dfrac{1}{20}$
(*c*) 2 (*d*) 20

142. If $a + b + c = 0$, $(a + b)(b + c)(c + a)$ equals **(M.C.A., 2005)**
(*a*) $ab(a + b)$ (*b*) $(a + b + c)^2$
(*c*) $-abc$ (*d*) $a^2 + b^2 + c^2$

143. If $a = 7$, $b = 5$, $c = 3$, then the value of $a^2 + b^2 + c^2 - ab - bc - ca$ is
(*a*) -12 (*b*) 0
(*c*) 8 (*d*) 12

144. Both addition and multiplication of numbers are operations which are
(*a*) neither commutative nor associative
(*b*) associative but not commutative
(*c*) commutative but not associative
(*d*) commutative and associative

145. Which of the following digits will replace the *H* marks in the following equation?
$9H + H8 + H6 = 230$
(*a*) 3 (*b*) 4
(*c*) 5 (*d*) 9
(*e*) None of these

146. Find the missing number in the following addition problem:

$$\begin{array}{r} 8\ 3\ 5 \\ 4\ *\ 8 \\ +\ 9\ *\ 4 \\ \hline 2\ 2\ *\ 7 \\ \hline \end{array}$$

(*a*) 0 (*b*) 4
(*c*) 6 (*d*) 9

147. What number should replace *M* in this multiplication problem?

$$\begin{array}{r} 3\ M\ 4 \\ \times\ 4 \\ \hline 1\ 2\ 1\ 6 \\ \hline \end{array}$$

(*a*) 0 (*b*) 2
(*c*) 4 (*d*) 8

148. If *p* and *q* represent digits, what is the maximum possible value of *q* in the statement **(S.S.C., 2010)**
$5p9 + 327 + 2q8 = 1114$?
(*a*) 6 (*b*) 7
(*c*) 8 (*d*) 9

149. What would be the maximum value of *Q* in the following equation?
$5P7 + 8Q9 + R32 = 1928$
(*a*) 6 (*b*) 8
(*c*) 9 (*d*) Data inadequate
(*e*) None of these

150. What should come in place of * mark in the following equation?
1*5$4 ÷ 148 = 78
(*a*) 1 (*b*) 4
(*c*) 6 (*d*) 8
(*e*) None of these

151. If 6*43 – 46@9 = 1904, which of the following should come in place of *?
(*a*) 4 (*b*) 6
(*c*) 9 (*d*) Cannot be determined
(*e*) None of these

152. What should be the maximum value of *Q* in the following equation?
$5P9 - 7Q2 + 9R6 = 823$
(*a*) 5 (*b*) 6
(*c*) 7 (*d*) 9
(*e*) None of these

153. In the following sum, '?' stands for which digit?
? + 1? + 2? +? 3 +? 1 = 21?
(*a*) 4 (*b*) 6
(*c*) 8 (*d*) 9
(*e*) None of these

Directions (Questions 154–155): *These questions are based on the following information:*
CBA + *CCA* = *ACD*, where *A*, *B*, *C* and *D* stand for distinct digits and *D* = 0.

154. B takes the value
(*a*) 0 (*b*) 5
(*c*) 9 (*d*) 0 or 9

155. C takes the value
(*a*) 0 (*b*) 2
(*c*) 2 or 3 (*d*) 5

156. A 3–digit number $4a3$ is added to another 3–digit number 984 to give the four-digit number $13b7$, which is divisible by 11. Then, $(a + b)$ is **(M.B.A., 2006)**
(*a*) 10 (*b*) 11
(*c*) 12 (*d*) 15

157. If $ab\overline{)252(}\,ba$, the values of a and b are **(I.A.M., 2007)**

```
ab)252(ba
   24
    12
    12
    ×
```

(*a*) 1, 2 (*b*) 2, 3
(*c*) 1, 3 (*d*) None of these

158.

```
        * * *
          × *
    -------
    8 * * 1
```

In the above multiplication problem, * is equal to
(*a*) 1 (*b*) 3
(*c*) 7 (*d*) 9

159. If * means adding 6 times the second number to the first number, then (1 * 2) * 3 equals
(*a*) 21 (*b*) 31
(*c*) 91 (*d*) 93

160. If $1 \times 2 \times 3 \times \ldots\ldots \times n$ is denoted by $\lfloor n$, then $\lfloor 8 - \lfloor 7 - \lfloor 6$ is equal to
(*a*) $6 \times 7 \times \lfloor 8$ (*b*) $7 \times 8 \times \lfloor 7$
(*c*) $6 \times 8 \times \lfloor 6$ (*d*) $7 \times 8 \times \lfloor 6$

161. The highest power of 9 dividing 99! completely is
(*a*) 11 (*b*) 20
(*c*) 22 (*d*) 24

162. For an integer n, $n! = n(n - 1)(n - 2)\ldots\ldots\ldots\ldots 3.2.1.$ **(P.C.S., 2008)**
Then, $1! + 2! + 3! + \ldots\ldots\ldots + 100!$ when divided by 5 leaves remainder
(*a*) 0 (*b*) 1
(*c*) 2 (*d*) 3

163. The number of prime factors in the expression $6^{10} \cdot 7^{17} \cdot 11^{27}$ is equal to
(*a*) 54 (*b*) 64
(*c*) 71 (*d*) 81

164. What is the number of prime factors contained in the product $30^7 \times 22^5 \times 34^{11}$?
(*a*) 49 (*b*) 51
(*c*) 52 (*d*) 53

165. What number multiplied by 48 will give the same product as 173 multiplied by 240?
(*a*) 495 (*b*) 545
(*c*) 685 (*d*) 865

166. A positive number, which when added to 1000, gives a sum which is greater than when it is multiplied by 1000. This positive integer is
(*a*) 1 (*b*) 3
(*c*) 5 (*d*) 7

167. 7 is added to a certain number ; the sum is multiplied by 5 ; the product is divided by 9 and 3 is subtracted from the quotient. Thus, if the remainder left is 12, what was the original number? **(S.S.C., 2005)**
(*a*) 20 (*b*) 30
(*c*) 40 (*d*) 60

168. Symbiosis runs a Corporate Training Programme. At the end of running the first programme, its total takings were ₹ 38950. There were more than 45 but less than 100 participants. What was the participant fee for the programme? **(SNAP, 2005)**
(*a*) ₹ 410 (*b*) ₹ 450
(*c*) ₹ 500 (*d*) ₹ 510

169. The sum of four consecutive even numbers A, B, C and D is 180. What is the sum of the set of next four consecutive even numbers? **(Bank Recruitment, 2008)**
(*a*) 196 (*b*) 204
(*c*) 212 (*d*) 214
(*e*) None of these

170. A young girl counted in the following way on the fingers of her left hand. She started calling the thumb 1, the index finger 2, middle finger 3, ring finger 4, little finger 5, then reversed direction, calling the ring finger 6, middle finger 7, index finger 8, thumb 9 and then back to the index figure for 10, middle finger for 11, and so on. She counted upto 1994. She ended on her
(*a*) thumb (*b*) index finger
(*c*) middle finger (*d*) ring finger

171. Given $n = 1 + x$ and x is the product of four consecutive integers. Then which of the following is true?
I. n is an odd integer. II. n is prime.
III. n is a perfect square.
(*a*) Only I is correct
(*b*) Only III is correct
(*c*) Both I and II are correct
(*d*) Both I and III are correct

172. If $x + y = 15$ and $xy = 56$, then what is the value of $x^2 + y^2$? **(L.I.C.A.D.O., 2007)**
(*a*) 110 (*b*) 113
(*c*) 121 (*d*) Cannot be determined
(*e*) None of these

173. Given that $(1^2 + 2^2 + 3^2 + \ldots + 20^2) = 2870$, the value of $(2^2 + 4^2 + 6^2 + \ldots + 40^2)$ is
(*a*) 2870 (*b*) 5740
(*c*) 11480 (*d*) 28700